

OS LAB

PRACTICAL – 7

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

memory.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <stdint.h>

int global_var = 100;
int uninitialized_global;

static int static_global = 200;

void display()
{
    printf("Inside display() function\n");
}

int main()
{
    int local_var = 10;
    static int static_local = 300;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
    }
}
```

```

    return 1;
}

*heap_var = 400;

printf("Process ID: %d\n", getpid());

printf("\n----- Memory Addresses ----- \n");

printf("Code address      : %p\n",
       (void *)(uintptr_t)display);

printf("Initialized global  : %p\n",
       (void *)&global_var);

printf("Uninitialized global : %p\n",
       (void *)&uninitialized_global);

printf("Static global       : %p\n",
       (void *)&static_global);

printf("Static local        : %p\n",
       (void *)&static_local);

printf("Heap variable        : %p\n",
       (void *)heap_var);

printf("Stack variable       : %p\n",
       (void *)&local_var);

printf("\nPress Enter to exit...\n");
getchar();

free(heap_var);

return 0;
}

```

```
lahari@DESKTOP-AIC33F7: ~/ OS_Lab
lahari@DESKTOP-AIC33F7:~/OS_Lab$ mkdir practical7
lahari@DESKTOP-AIC33F7:~/OS_Lab$ cd practical
-bash: cd: practical: No such file or directory
lahari@DESKTOP-AIC33F7:~/OS_Lab$ cd practical7
lahari@DESKTOP-AIC33F7:~/OS_Lab/practical7$ nano memory.c
lahari@DESKTOP-AIC33F7:~/OS_Lab/practical7$ gcc -Wall -Wextra -O 0 -g memory.c -o memory
/usr/bin/ld: cannot find 0: No such file or directory
collect2: error: ld returned 1 exit status
lahari@DESKTOP-AIC33F7:~/OS_Lab/practical7$ gcc -Wall -Wextra -O0 -g memory.c -o memory
lahari@DESKTOP-AIC33F7:~/OS_Lab/practical7$ ./memory
Process ID: 585

----- Memory Addresses -----
Code address      : 0x597ae6d14209
Initialized global : 0x597ae6d17010
Uninitialized global : 0x597ae6d17020
Static global     : 0x597ae6d17014
Static local      : 0x597ae6d17018
Heap variable     : 0x597b25c972a0
Stack variable    : 0x7ffdaf00c65c

Press Enter to exit...

lahari@DESKTOP-AIC33F7:~/OS_Lab/practical7$ ./memory
Process ID: 590

----- Memory Addresses -----
Code address      : 0x5d4b9a33f209
Initialized global : 0x5d4b9a342010
Uninitialized global : 0x5d4b9a342020
Static global     : 0x5d4b9a342014
Static local      : 0x5d4b9a342018
Heap variable     : 0x5d4bafb162a0
Stack variable    : 0x7fff914dccac
```